
2019-Ph-H1-Q8

Section: Electricity

Topic: Capacitors

Question:

A 100 μF capacitor is charged to 10 V. What is the energy stored?

Answer:

Use $E = \frac{1}{2}CV^2$

Convert 100 μF to farads: $100 \times 10^{-6} \text{ F}$

$$E = 0.5 \times 100 \times 10^{-6} \times 10^2 = 0.5 \times 100 \times 10^{-6} \times 100 = 0.005 \text{ J}$$

Correct Option: C (0.05 J)

Guidance for Students:

Always convert μF to F! Many students forget this. Check your units before calculating.

Revision Tips:

- $E = \frac{1}{2}CV^2$
 - $1 \mu\text{F} = 10^{-6} \text{ F}$
 - Check calculator input carefully — especially powers of ten!
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